

BRIAN ZHOU LIU

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EDUCATION

University of California, San Diego

B.S. in Data Science (AI & ML Specialization)

La Jolla, California

GPA: 3.95/4.00 | Expected June 2028

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript/JavaScript, Rust, Bash

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, React, FastAPI

Data & Infrastructure: PostgreSQL, MongoDB, ChromaDB, Git, Docker, GCP (Cloud Run, Cloud SQL)

AI & Agents: Claude Code, Gemini API, retrieval-augmented generation (RAG), Dialogflow CX

EXPERIENCE

Q-Lab, UC San Diego

Research Intern (Advisor: Prof. Lianhui Qin)

La Jolla, California

January 2026 - Present

- Co-developed SIGA, a Claude Code plugin that automates setup for scientific simulations; implemented its ChromaDB search, XML validator, and plugins; co-authored a paper accepted as a NeurIPS 2026 poster ([arXiv:2606.09774](https://arxiv.org/abs/2606.09774)).
- Led a 30-task OpenFOAM simulation benchmark; SIGA completed all cases and reached 87% accuracy in its best setup.
- Tested SIGA on new GEOS simulation tasks; accuracy rose from 72.0% to 78.9% and run-to-run variation fell about 16x.
- Designed CellShift, a seven-dataset benchmark testing models on cell fates and experimental conditions excluded from training; prepared data and evaluated models including CellTempo (ICLR 2027 submission).

Asakana (YC F26)

Product Development Intern

Remote

October 2025 - February 2026

- Deployed a Gemini Flash-Lite OCR/ETL pipeline to extract supplier data for 10,000+ products from PDF/Excel sheets and load structured records into MongoDB.
- Developed a Dialogflow CX agent for customer orders, connected to Cloud SQL and REST APIs, with Twilio SMS notifications and automatic pricing-rule enforcement.

Rare AI Lab, UC San Diego

Research Intern (Advisor: Prof. Aobo Li)

La Jolla, California

September 2024 - December 2025

- Implemented a multi-fidelity Rare Event Surrogate Model combining conditional neural processes and Gaussian processes, reducing detector-simulation cost by 90%.
- Co-first-authored a [NeurIPS 2025 Machine Learning and the Physical Sciences workshop paper](#) on detector design optimization.

MAIX Lab, Emory University

Research Intern (Advisor: Prof. Ran Xiao) | Regeneron STS Top 300 Scholar

Atlanta, Georgia

March 2023 - November 2023

- Developed anatomically informed XResNet features and cross-lead 1D convolutions for electrocardiogram screening, achieving 93.7% AUC and 85.5% sensitivity.

SELECTED PROJECTS

FlashMath: AI-Assisted Math Study Desktop App

Tauri, Rust, React, TypeScript

Independent Project | [GitHub](#)

2026

- Developed a Tauri desktop app with Rust/React for personal math study, with local persistence, screenshot/PDF ingestion, LaTeX rendering, and provider-agnostic AI OCR.
- Adapted SM-2 spaced repetition to use time spent per review and schedule more frequent practice near exam deadlines.

Blueprint: Codebase Summarization & Vulnerability Detection

NexHacks | 1st Place DevTools | 3rd Place Token Company | Best Use of Gemini API

Pittsburgh, Pennsylvania | [Devpost](#)

January 2026

- Parsed source code into abstract syntax trees (ASTs) to visualize dependencies with React Flow/D3.js in VS Code.
- Integrated Gemini, parallel AI agents, and bear-1 token compression for code analysis; tested on MongoDB's codebase.

CARP: Health Monitoring & Catch Optimization System

Sushi Hackathon at Stanford | 3rd Place

Stanford, California | [Slides](#)

October 2025

- Prototyped an Arduino wrist brace with FSR-402 pressure sensors and Bluetooth; integrated a Random Forest using clinical features to predict carpal tunnel syndrome risk in a Python dashboard.
- Developed a full-stack dashboard with a retrieval-augmented market-insights agent and geospatial fishing maps covering 50,000+ data points across 5,700+ species.

CiteTrace: Research Knowledge Graph for Insight Discovery

ACM x Intel Hackathon | 1st Place

Santa Clara, California | [Devpost](#)

May 2025

- Created a React Native/Expo paper graph with SVG rendering, D3 convex hulls, and similarity-weighted edges.
- Implemented union-find clustering over paper similarity scores to group related research and visualize clusters.
- Implemented a retrieval-augmented research agent that answers questions over papers and returns cited passages.